

✓ Colab Training Notebook for spiking-transformer-cs4782

This notebook sets up a Colab GPU runtime, prepares enwik8, and runs training.

```
from google.colab import drive
drive.mount('/content/drive')
```

```
!cp -r /content/drive/MyDrive/DeepLearning/spikegpt/spiking-transformer-cs4782 /content/
%cd /content/spiking-transformer-cs4782
```

Drive already mounted at /content/drive; to attempt to forcibly remount, call drive.mount("/content/drive", force_remount=True).

```
!pip install -r requirements.txt
!pip install spikingjelly
```

Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.7->)

Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from sympy>=1.13.
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from jinja2->torch->

```
import torch
print("torch:", torch.__version__)
print("cuda available:", torch.cuda.is_available())
if torch.cuda.is_available():
    print("gpu:", torch.cuda.get_device_name(0))
```

```
torch: 2.10.0+cu128
cuda available: True
gpu: Tesla T4
```

```
!wget http://mattmahoney.net/dc/enwik8.zip
!unzip -o enwik8.zip
```

```
--2026-04-17 05:11:49--  http://mattmahoney.net/dc/enwik8.zip
Resolving mattmahoney.net (mattmahoney.net)... 20.119.76.151
Connecting to mattmahoney.net (mattmahoney.net)|20.119.76.151|:80... connected.
HTTP request sent, awaiting response... 200 OK
Length: 36445475 (35M) [application/zip]
Saving to: 'enwik8.zip.3'
```

```
enwik8.zip.3          100%[=====>]  34.76M  8.58MB/s   in 4.0s
```

```
2026-04-17 05:11:54 (8.58 MB/s) - 'enwik8.zip.3' saved [36445475/36445475]
```

```
Archive: enwik8.zip
  inflating: enwik8
```

```
from pathlib import Path

raw = Path("enwik8").read_bytes()

out_dir = Path("data/enwik8_split")
out_dir.mkdir(parents=True, exist_ok=True)

n = len(raw)
train_end = int(0.9 * n)
valid_end = int(0.95 * n)

(out_dir / "train.txt").write_bytes(raw[:train_end])
(out_dir / "valid.txt").write_bytes(raw[train_end:valid_end])
(out_dir / "test.txt").write_bytes(raw[valid_end:])

print("done")
print("train:", (out_dir / "train.txt").stat().st_size)
print("valid:", (out_dir / "valid.txt").stat().st_size)
print("test:", (out_dir / "test.txt").stat().st_size)
```

```
done
train: 90000000
valid: 5000000
test: 5000000
```

```
# cell to update all the code if you change it
import sys
import importlib
from pathlib import Path

code_path = str(Path("code").resolve())
if code_path not in sys.path:
    sys.path.insert(0, code_path)

for name in ["model", "config", "dataset", "train", "test", "generate"]:
    if name in sys.modules:
        importlib.reload(sys.modules[name])

import model
import config
```

```
import dataset
import train
import test
import generate

from model import SpikingGPT
from config import GPTConfig, TrainerConfig
from dataset import Enwik8Dataset

print("project modules refreshed")
```

project modules refreshed

```
# test before actual training
import sys
from pathlib import Path
import torch

sys.path.insert(0, str(Path("code").resolve()))

from config import GPTConfig
from model import SpikingGPT
from dataset import Enwik8Dataset

device = "cuda" if torch.cuda.is_available() else "cpu"
print("device:", device)

cfg = GPTConfig(ctx_len=64, n_embd=64, n_layer=2)

dataset = Enwik8Dataset("data/enwik8_split/train.txt", cfg.ctx_len)
x, y = dataset[0]
x = x.unsqueeze(0).to(device)
y = y.unsqueeze(0).to(device)

model = SpikingGPT(cfg).to(device)

print("x shape:", x.shape)
print("y shape:", y.shape)

loss = model(x, y)
print("smoke test loss:", loss.item())
```

device: cuda
x shape: torch.Size([1, 64])
y shape: torch.Size([1, 64])
smoke test loss: 5.620720863342285

```
!ls data/enwik8_split
!python code/train.py
```

```
test.txt train.txt valid.txt
using device: cuda
smoke test loss: 5.7032084465026855
epoch 1 step 0 loss 5.7119
epoch 1 step 50 loss 3.1407
epoch 1 step 100 loss 2.6646
epoch 1 step 150 loss 2.7450
epoch 1 step 200 loss 2.7132
epoch 1 step 250 loss 2.2252
epoch 1 step 300 loss 2.2530
epoch 1 step 350 loss 2.1477
epoch 1 step 400 loss 2.1684
epoch 1 step 450 loss 1.7992
epoch 1 step 500 loss 2.3807
epoch 1 step 550 loss 2.2185
epoch 1 step 600 loss 2.2081
epoch 1 step 650 loss 1.8971
epoch 1 step 700 loss 2.0893
epoch 1 step 750 loss 2.0513
epoch 1 step 800 loss 1.8885
epoch 1 step 850 loss 1.8564
epoch 1 step 900 loss 1.8422
```

```
epoch 1 step 950 loss 1.8138
epoch 1 step 1000 loss 1.9472
epoch 1 step 1050 loss 1.9321
epoch 1 step 1100 loss 1.5101
epoch 1 step 1150 loss 1.9460
epoch 1: train_loss=2.2349 val_loss=1.8223
```

```
!mkdir -p /content/drive/MyDrive/spiking-transformer-cs4782-results
!cp -r results /content/drive/MyDrive/spiking-transformer-cs4782-results/
```

```
!python code/test.py
```

```
available checkpoints:
- epoch_1.pt
using checkpoint: epoch_1.pt
test_loss=2.1212
test_perplexity=8.3413
test_bpc=3.0603
R_hat=0.187486
params_m=3.5512
```

Energy Table Inputs

```
T=256
d=256
R_hat=0.187486
E_MAC=4.5 pJ
E_AC=0.9 pJ
```

Energy Table Values

```
Q/R,K,V: vanilla = E_MAC * 3*T*d^2 = 4.5 * 3 * 256 * 256^2 = 2.2649e+08 pJ
Q/R,K,V: spikegpt = E_AC * R_hat * 3*T*d^2 = 0.9 * 0.187486 * 3 * 256 * 256^2 = 8.4928e+06 pJ
f(Q/R,K,V): vanilla = E_MAC * 2*T^2*d = 4.5 * 2 * 256^2 * 256 = 1.5099e+08 pJ
f(Q/R,K,V): spikegpt = E_MAC * 7*T*d = 4.5 * 7 * 256 * 256 = 2.0644e+06 pJ
Scale: vanilla = E_MAC * T^2 = 4.5 * 256^2 = 2.9491e+05 pJ
Softmax: vanilla = E_MAC * 2*T^2 = 4.5 * 2 * 256^2 = 5.8982e+05 pJ
FFN layer 1: vanilla = E_MAC * FL_MLP1 = 4.5 * (256 * 1024) = 1.1796e+06 pJ
FFN layer 1: spikegpt = E_AC * R_hat * FL_MLP1 = 0.9 * 0.187486 * (256 * 1024) = 4.4233e+04 pJ
FFN layer 2: vanilla = E_MAC * FL_MLP2 = 4.5 * (256 * 256) = 2.9491e+05 pJ
FFN layer 2: spikegpt = E_AC * R_hat * FL_MLP2 = 0.9 * 0.187486 * (256 * 256) = 1.1058e+04 pJ
FFN layer 3: vanilla = E_MAC * FL_MLP3 = 4.5 * (1024 * 256) = 1.1796e+06 pJ
FFN layer 3: spikegpt = E_AC * R_hat * FL_MLP3 = 0.9 * 0.187486 * (1024 * 256) = 4.4233e+04 pJ
Overall vanilla = 3.8103e+08 pJ
Overall spikegpt = 1.0657e+07 pJ
Overall ratio V/S = 35.7545x
```

Table-ready row

Method: SpikeGPT small reimplementation

Spiking: yes

L: 4

d: 256

T: 256

Train BPC: 4.1570

Test BPC: 3.0603

Complexity: $O(T \cdot d)$

Params (M): 3.5512

```
!python code/generate_colab.py
```

```
using device: cuda
using checkpoint: epoch_1.pt
ctx_len=256 n_embd=256 n_layer=4
```

prompt:

'The '

generated:

The same and such and new it to the gave and which a muring of the orden and the the from the creach such to mos

